

WEEKLY PROJECT PLAN (06 Oct 2025 – 20 Apr 2026)

PHASE 1 — Proposal & Foundation (6 Oct – 10 Nov 2025)

Goal: Establish theoretical foundation, validate research gap, and complete proposal submission.

Week	Dates	Focus & Tasks	Expected Outputs
W1	6–12 Oct	Define final project scope and aims (framework, not standalone app). Conduct a deep literature overview using Heidari et al. (2023) and Gong & Li (2024) to map detection methods (CNN, Transformer, multimodal, biological). Identify 3–4 subproblems: dataset bias, multimodal fusion, explainability, real-time deployment.	Project scope statement, subproblem table, annotated bibliography (10+ key papers).
W2	13–19 Oct	Detailed review of datasets (FaceForensics++, DFDC, KoDF, WildDeepfake, FakeAVCeleb). Compare dataset properties, diversity, realism (Kwon et al. 2021; Zi et al. 2020; Khalid et al. 2021). Begin initial draft of Problem Domain section.	Dataset comparison matrix, problem domain draft.
W3	20–26 Oct	Research baseline models from Pan et al. (2020) and Ahmed et al. (2022). Sketch high-level architecture: CNN + transformer hybrid. Draft Methodology and Evaluation Plan sections.	Initial proposal draft, architecture diagram.
W4	27 Oct – 2 Nov	Finalize Introduction, Problem Domain, Methodology, and Evaluation. Proofread and format proposal.	Proposal v2 for supervisor feedback.
W5	3–9 Nov	Incorporate supervisor comments. Final checks: citations (Harvard), references, ethics summary.	Final Proposal

PHASE 2 — Contextual Report & System Design (10 Nov 2025 – 12 Jan 2026)

Goal: Deepen research, design the system architecture, complete contextual report.

Week	Dates	Focus & Tasks	Expected Outputs
W6	10–16 Nov	Prepare full ethics pack (PIS, consent forms, questionnaire). Submit to Faculty Ethics Committee.	Ethics submission.
W7	17–23 Nov	Literature deep-dive: dataset realism & cross-dataset generalization issues (Kwon et al., Zi	Contextual Report draft –

		et al.). Start contextual report (literature review section).	Section 1 (Intro & Lit Review).
W8	24–30 Nov	Requirements engineering: functional, non-functional, ethical requirements. Use UML (use cases, class diagrams).	Requirements specification document.
W9	1–7 Dec	Design detailed system architecture: modular backend, detection module, frontend UI. Define APIs and data flow diagrams. Reference Lyu (2020) for explainability design.	Design diagrams and architecture specification.
W10	8–14 Dec	Implement small baseline CNN model (VGG-Face or MobileNet) on FaceForensics++ subset (Pan et al. 2020). Document experiment results.	Baseline model test report.
W11	15–21 Dec	Draft Methodology and Implementation Plan sections of contextual report. Incorporate journal-based justification (Ahmed et al. 2022; Heidari et al. 2023).	Contextual Report v1.
W12	22–28 Dec	Light week (holiday buffer). Format contextual report and prepare for submission.	Polished draft.
W13	29 Dec – 12 Jan	Final proofread and supervisor review. Integrate all diagrams, tables, references.	Submit Contextual Report.

PHASE 3 — Implementation, Testing & Evaluation (13 Jan – 22 Mar 2026)

Goal: Build, test, and refine the detection framework and evaluation pipeline.

Week	Dates	Focus & Tasks	Expected Outputs
W14	13–19 Jan	Build full data-preprocessing pipeline: frame extraction, audio segmentation, MFCC feature generation. Dataset integration (KoDF + FakeAVCeleb).	Preprocessing scripts, verified data samples.
W15	20–26 Jan	Train and evaluate visual CNN (Xception / MobileNet). Measure dataset-specific performance (FaceForensics++).	Visual model results (Accuracy, F1, AUC).
W16	27 Jan – 2 Feb	Implement audio branch using MFCC features and ResNet model (Khalid et al. 2021). Evaluate audio-only performance.	Audio model results, confusion matrix.
W17	3–9 Feb	Implement fusion (multi-modal) detector combining audio and video features. Begin API integration.	Multimodal model v1, feature fusion logs.

W18	10–16 Feb	Conduct internal testing, optimize parameters. Begin pilot usability study with 5 users. Collect qualitative feedback.	Pilot usability notes, model tuning.
W19	17–23 Feb	Apply explainability (Grad-CAM) for heatmaps to visualize fake regions (Lyu 2020). Evaluate cross-dataset generalization (KoDF → WildDeepfake).	Explainability visuals, cross-test accuracy chart.
W20	24 Feb – 2 Mar	Conduct main user study (~30 participants). Collect SUS data and user trust metrics (Sippy et al. 2024).	Raw user data, anonymized survey responses.
W21	3–9 Mar	Analyse survey results: mean SUS, qualitative comments. Statistical analysis (t-tests, mean comparisons).	Evaluation tables and graphs.
W22	10–16 Mar	Final optimization and code refactoring. Document APIs, backend integration, and system testing.	Finalized framework and documentation.
W23	17–23 Mar	Supervisor demo; fix minor bugs and performance issues.	Stable build v2.0.

PHASE 4 — Evaluation, Final Report & Submission (24 Mar – 20 Apr 2026)

Goal: Consolidate results, finalize report, prepare for submission and presentation.

Week	Dates	Focus & Tasks	Expected Outputs
W24	24–30 Mar	Compile evaluation data (accuracy, F1, SUS, latency). Generate visual results: ROC curves, heatmaps, confusion matrices.	Evaluation visuals and tables.
W25	31 Mar – 6 Apr	Write Results and Evaluation sections of final report. Integrate findings with literature discussion (link to Gong & Li 2024; Heidari 2023).	Report draft v1.
W26	7–13 Apr	Write Discussion, Conclusion, Future Work. Proofread for clarity and structure.	Full report v2.
W27	14–17 Apr	Prepare final presentation slides and demo. Final supervisor review and revisions.	Final report + presentation ready.
W28	18–19 Apr	Submit final report, source code, and demo video. Backup all materials.	Submission completed.
W29	20 Apr	Final deadline – Project submission day	Project officially submitted.